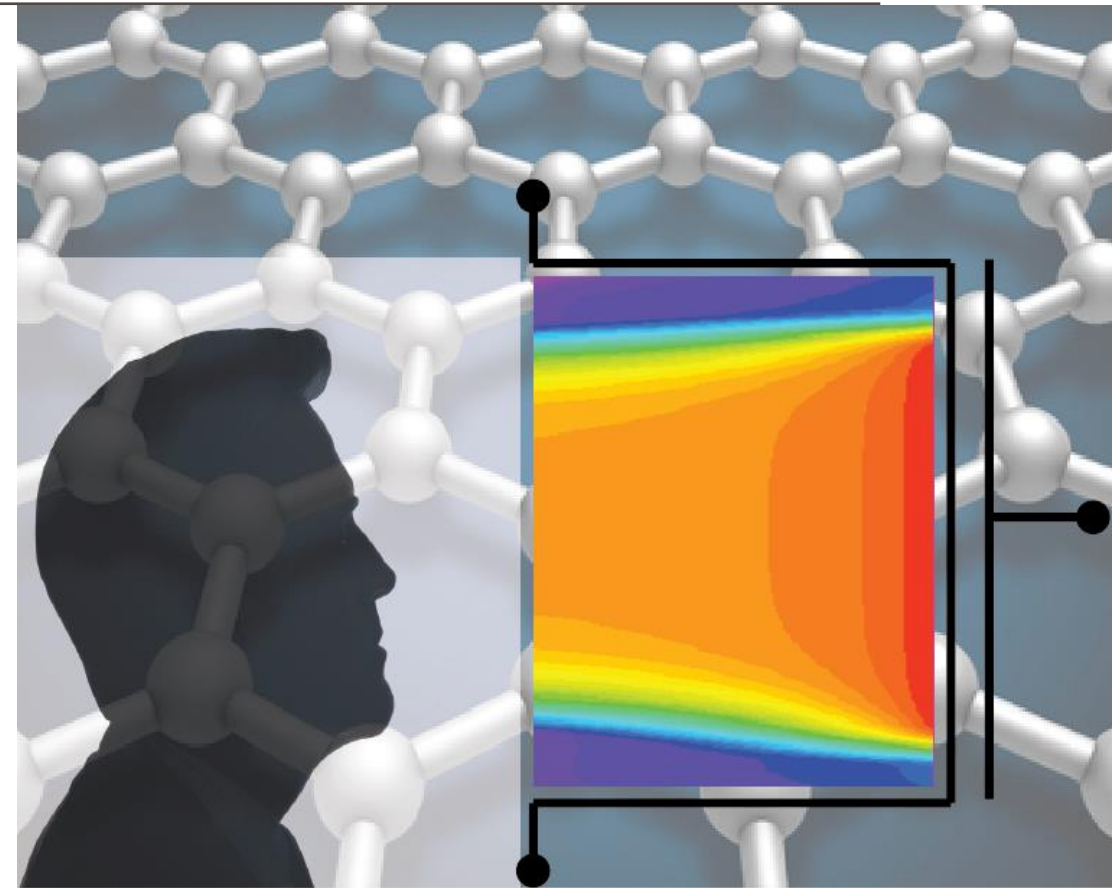


ADVANCED SOLID-STATE DEVICES

Week-7, Lecture-1

Sneh Saurabh
17th February, 2025



Field-Effect Transistors

Field-Effect Transistors



MOSFET

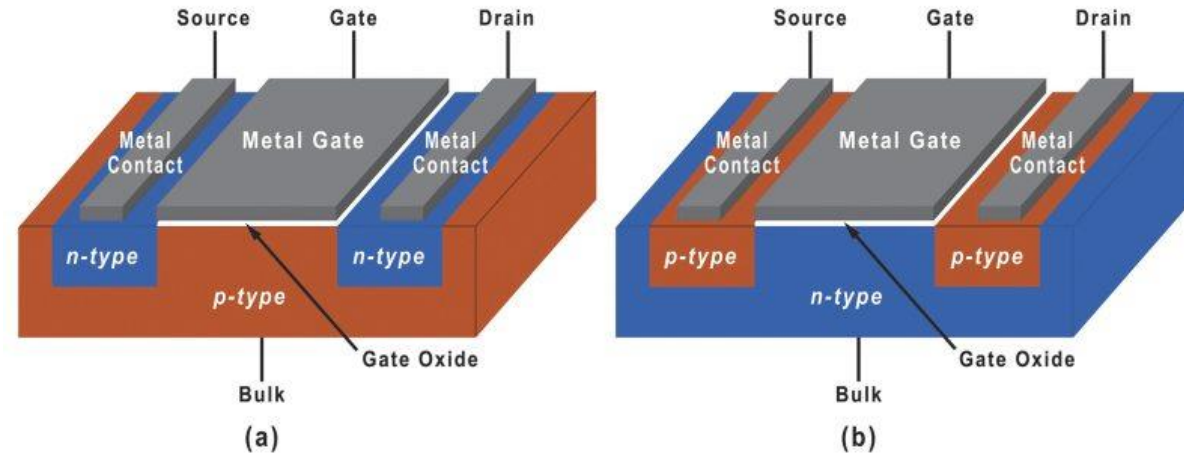
MOSFET: Structure

Structure: M-O-S

- Metal: Gate Terminal, can be highly doped polycrystalline silicon or Metal
- Oxide: Insulator (dielectric), SiO_2 , HfO_2 , also called MISFET
- Semiconductor: p-type or n-type Si

Other Terminals (Contact):

- Source/Drain: Ohmic
- Body/Substrate/Bulk: contact to substrate

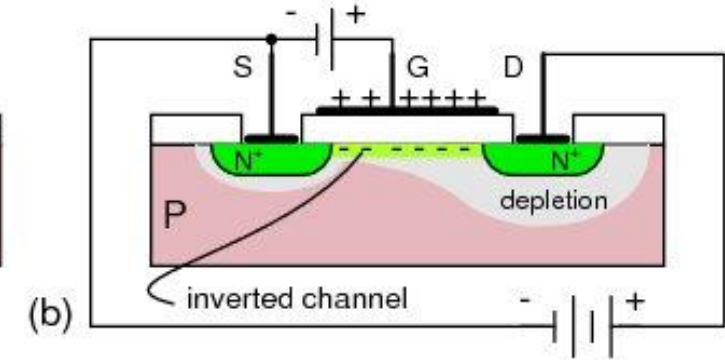
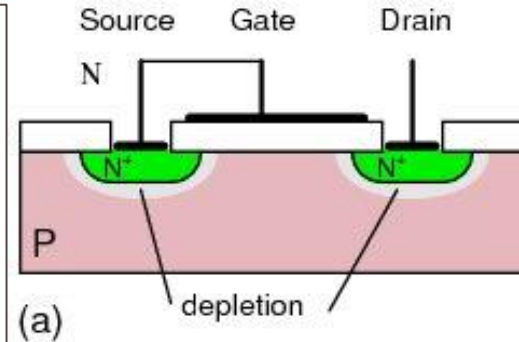


- Could not be realized due to the large density of surface states on the semiconductor surface
- Breakthrough came in early 1960s when excellent Si/SiO_2 interface could be realized using thermal oxidation
 - Allowed field-effect operations

MOSFET: Gate Control

No positive voltage to gate:

- Semiconductor is non-conducting
- Drain current zero



Small positive gate voltage:

- Conducting channel formed, channel works as resistor
- Current flows between source and drain

Large positive gate voltage:

- Channel resistance further decreases
- Slope of drain current vs. drain voltage becomes steeper

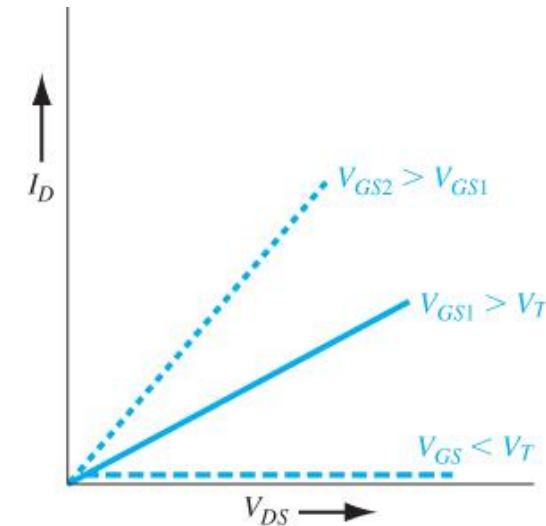


Figure 10.38 | I_D versus V_{DS} characteristics for small values of V_{DS} at three V_{GS} voltages.

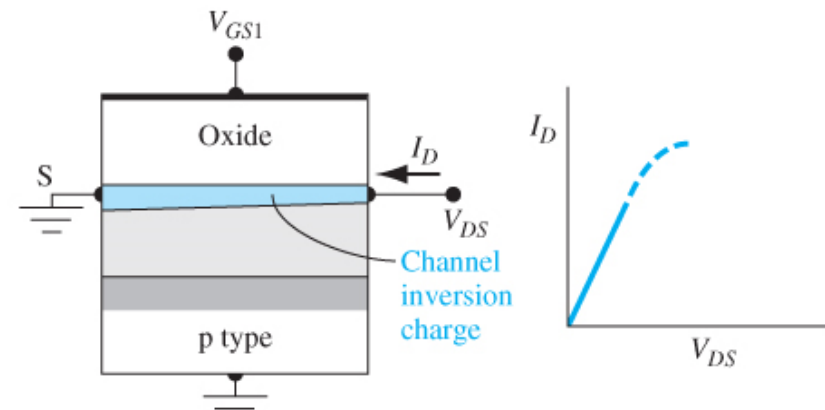
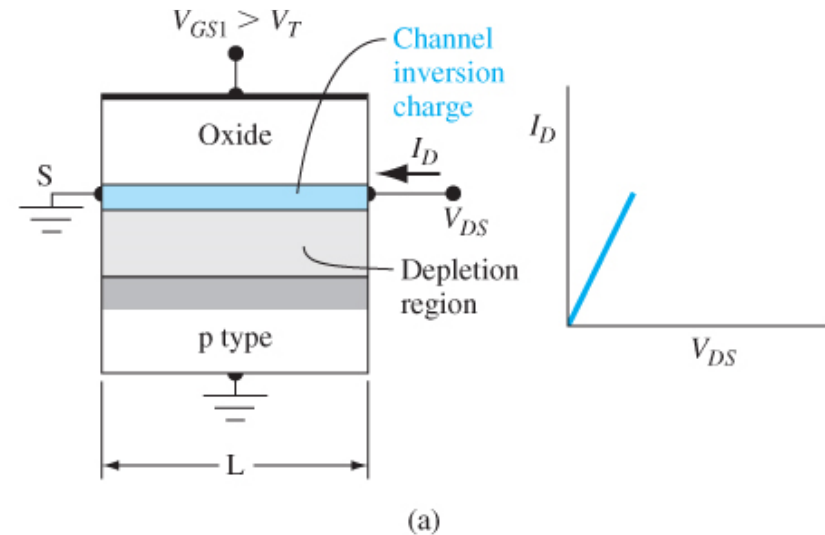
MOSFET: Drain Control (1)

Small drain voltage (gate voltage greater than threshold):

- Channel is like a resistor
- Linear drain current with drain voltage

Increased drain voltage (gate voltage greater than threshold):

- PN junction more reverse-biased on the drain side
- Channel becomes thinner on the drain side
- Effective channel resistance increases, slope decreases



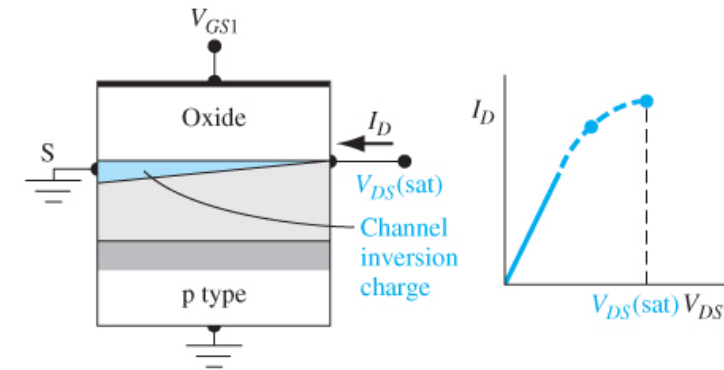
MOSFET: Drain Control (2)

Large drain voltage (gate voltage greater than threshold):

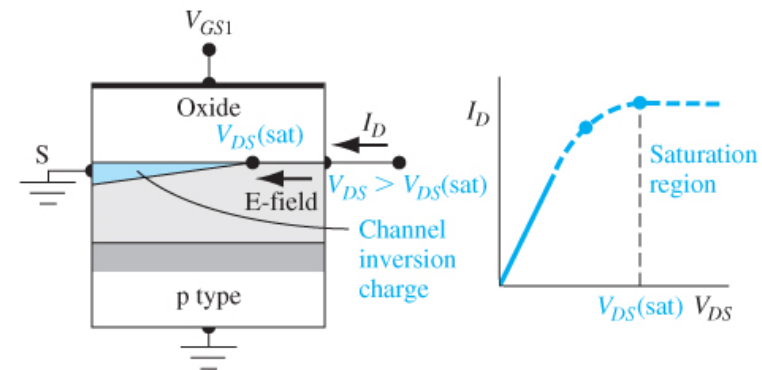
- Channel gets **pinched off** near the drain (but still conducting on source side)
- Further increase in resistance

Further increase in drain voltage (gate voltage greater than threshold):

- Increased voltage do not increase current (simply moves pinch off point toward source), drain current **saturates**
- Extra voltage appears across the pinched off region (that has higher resistance)



(c)



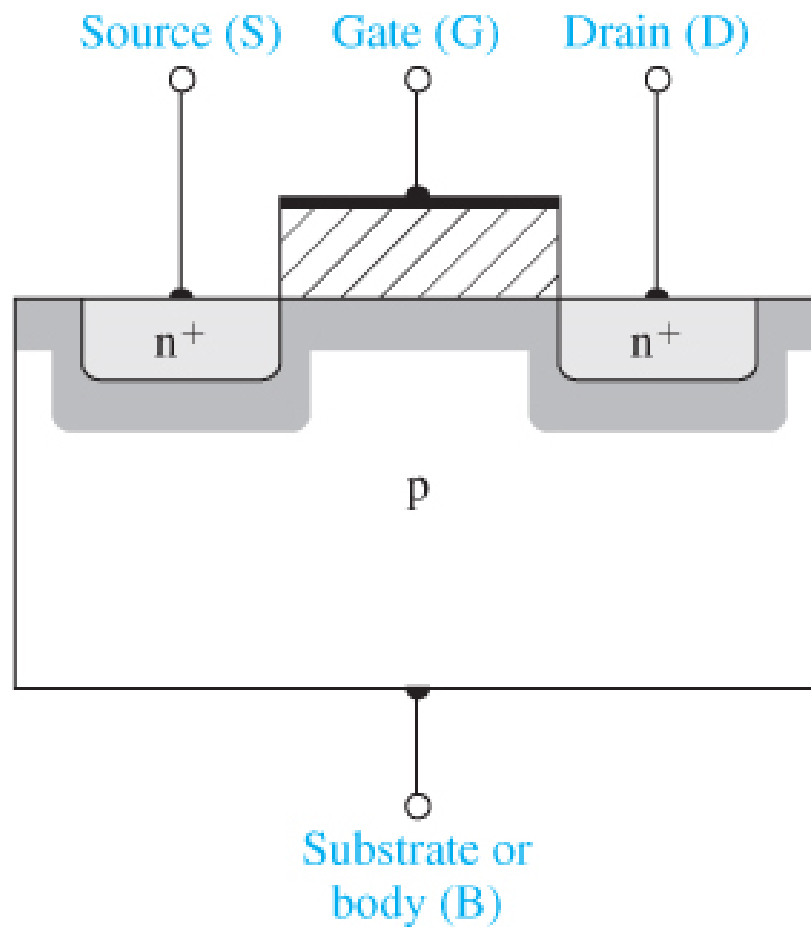
(d)

FET: Summary

	JFET	MESFET	MOSFET/MISFET
Operating principle	Field-effect	Field-effect	Field-effect
Current Carrier	Majority carrier	Majority carrier	Majority carrier
Gate Isolation	Reverse-biased PN junction	Reverse-biased Schottky junction	Insulator/Oxide
Gate Current	Lower	Low	Lowest
Applications	Less used	Less used	Workhorse of CMOS

MOSFETs

MOS Capacitor



- Operation of MOSFET primarily decided by the behavior of the MOS capacitor

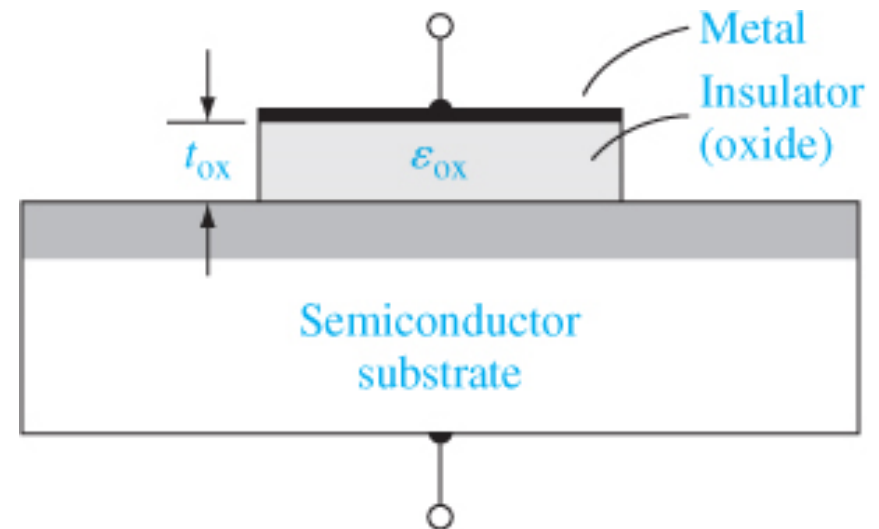
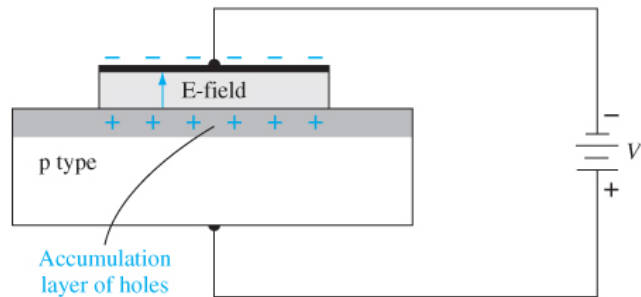
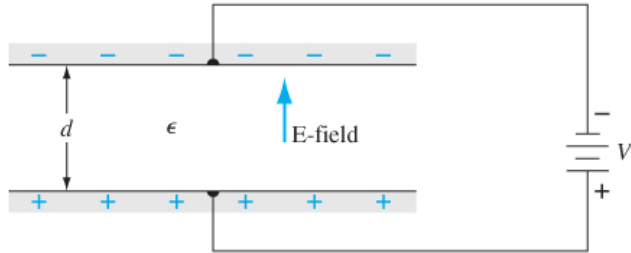


Figure 10.1 | The basic MOS capacitor structure.

Parallel Plate vs. MOS Capacitor (1)

- Parallel plate capacitor (with dielectric) and MOS capacitor have several things in common



- Application of voltage leads to accumulation of charge on the plates of the parallel plate capacitor
- Induced electric field in the dielectric
- Application of -ve voltage to metal leads to accumulation of -ve charge on metal plate of MOS capacitor
- Induced electric field penetrates the semiconductor
- Holes get accumulated at the oxide-semiconductor junction due to force of electric field
- Accumulated charge at oxide-semiconductor junction is similar to positive plate of the capacitor

Parallel Plate vs. MOS Capacitor (2)

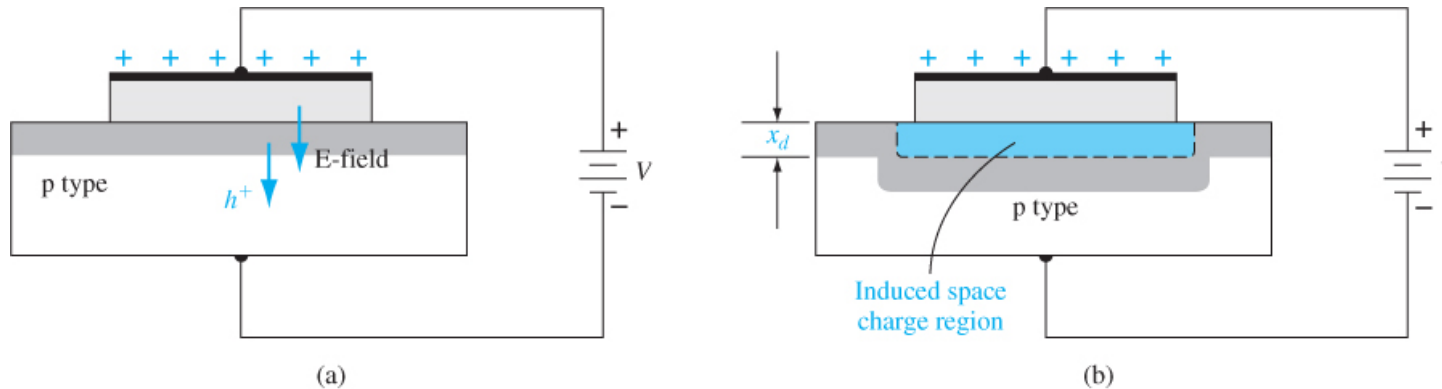


Figure 10.3 | The MOS capacitor with a moderate positive gate bias, showing (a) the electric field and charge flow and (b) the induced space charge region.

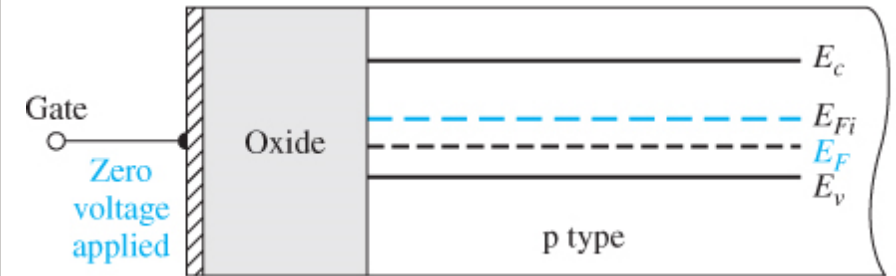
- Application of +ve voltage to metal leads to accumulation of +ve charge on metal plate of the MOS capacitor
- Induced electric field penetrates the semiconductor

- Holes move away from the oxide-semiconductor junction due to force of electric field
- Depletion region created at oxide-semiconductor junction having fixed -ve ionized acceptor atoms
- Depletion region having -ve ionized acceptor works as plate with -ve charge

Band Diagram for Ideal Case (1)

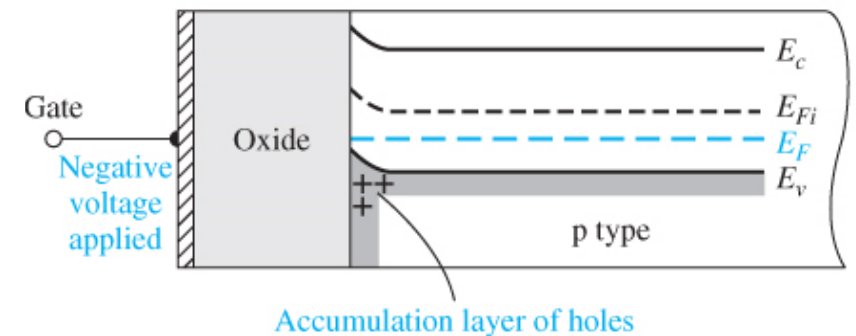
Zero voltage to Gate:

- Assume that when the bands are flat for a **p-type semiconductor**
- Actually, flat band will depend on the work function of metal and semiconductor (we are assuming ideal condition for now)
- This is the *flat band condition*



Negative voltage to Gate:

- Band moves upward at the oxide-semiconductor interface
- Fermi level constant (equilibrium condition exists, no current flows through oxide)
- Fermi level close to valence band near interface, accumulation of holes: *accumulation mode*



Band Diagram for Ideal Case (2)

Positive voltage to Gate:

- Bands move downward near the interface
- Fermi level close to intrinsic level
- Depletion region created (width x_d)
- MOS capacitor in *depletion mode*

Large positive voltage to Gate:

- More band bending and larger depletion region
- Conduction band close to the Fermi level near interface
- Surface becomes n-type (inversion layer created)
- MOS capacitor in inversion mode

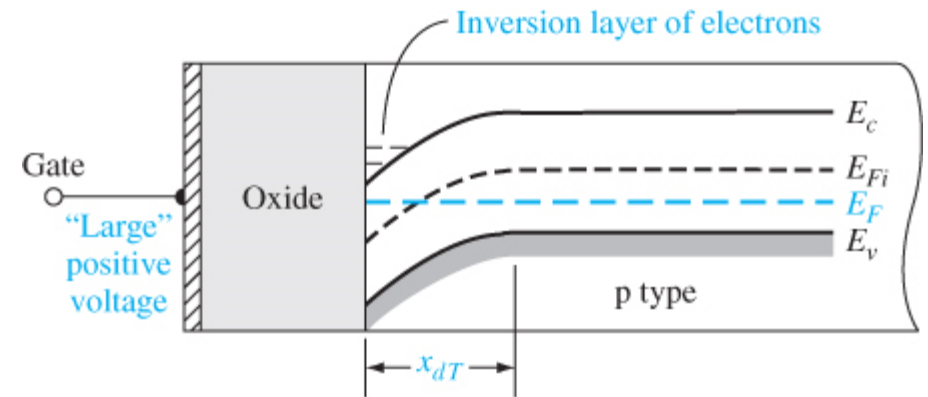
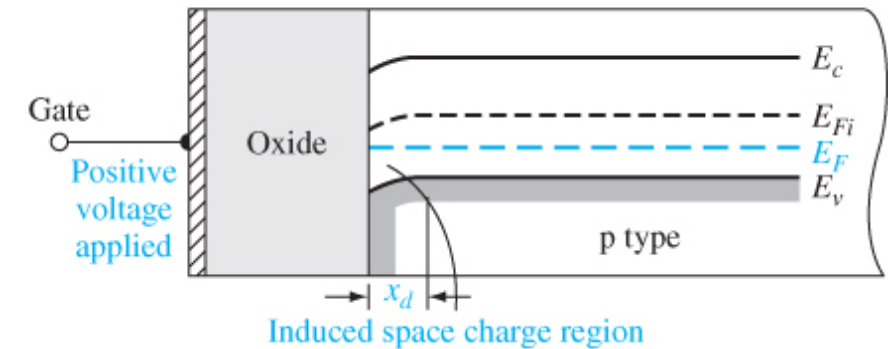
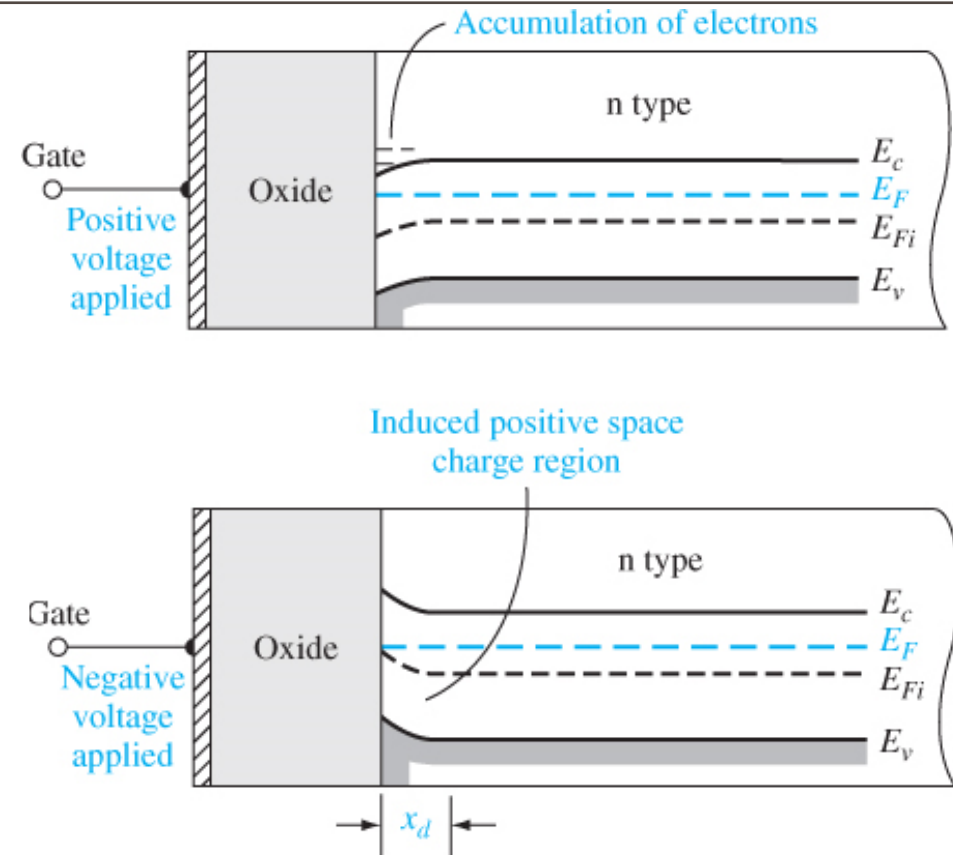
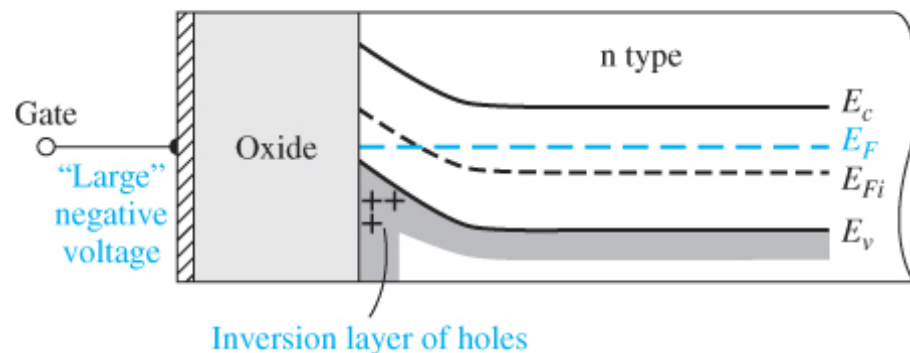


Figure 10.5 | The energy-band diagram of the MOS capacitor with a p-type substrate for a "large" positive gate bias.

Band Diagram for Ideal Case: N-type

- Similar behavior observed for n-type silicon
- Gate voltage polarity is opposite compared to p-type
- Accumulation of electrons on positive bias
- Depletion on small negative bias
- Inversion at surface on large negative bias (inversion layer consists of holes)



MOSFETs

MOS Capacitor: Surface Potential

- When voltage is applied to the gate, charges appear in the semiconductor
- Practically, all charges reside within a small region adjacent to the surface
- Outside the surface region, the semiconductor is practically neutral

Surface potential definition:

- Total potential drop ψ_s across the region, defined from the *surface* to the point in the *bulk outside this region*

NOTE: In Neaman's book surface potential is denoted as ϕ_s [We do not use this notation since it can be confused with work function of semiconductor]

- Surface potential is an important physical quantity to model and characterize the MOSFET behavior

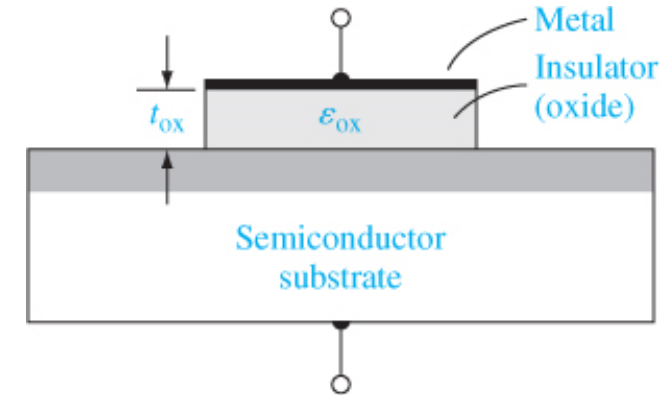


Figure 10.1 | The basic MOS capacitor structure.

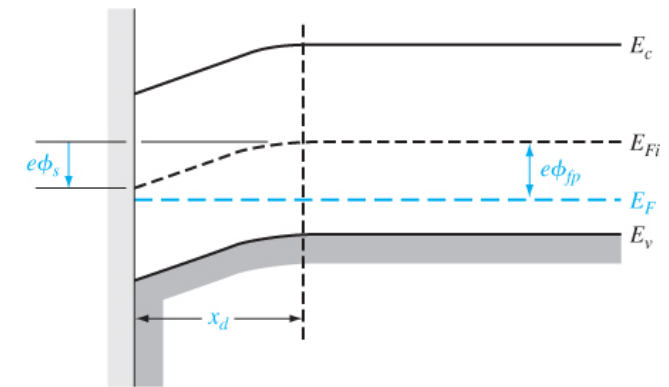


Figure 10.8 | The energy-band diagram in the p-type semiconductor, indicating surface potential.

MOS Capacitor: Surface Potential vs. Electron Concentration at Surface (1)

Problem: Assume that the surface potential is ψ_s . The acceptor concentration is N_A in the bulk. The difference in the Fermi level and the intrinsic energy level in the bulk is $\phi_{Fp} = E_{Fi} - E_F$.

1. Express N_A as a function of n_i and ϕ_{Fp} .

Then, re-write n_i as a function of N_A and ϕ_{Fp}

2. Express surface concentration of electron $n_{surface}$ as a function of ψ_s and ϕ_{Fp} .

3. Express concentration of electron $n_{surface}$ as a function of ψ_s , ϕ_{Fp} and N_A .

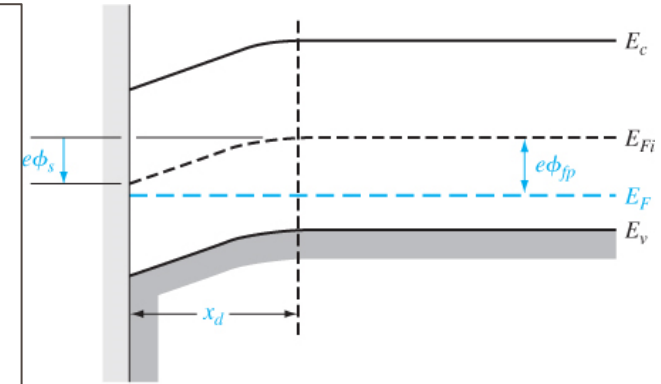


Figure 10.8 | The energy-band diagram in the p-type semiconductor, indicating surface potential.

$$1. \quad p = N_A = n_i \exp\left(\frac{E_{Fi} - E_F}{kT}\right) = n_i \exp\left(\frac{\phi_{Fp}}{kT/q}\right)$$

$$\Rightarrow n_i = N_A \exp\left(-\frac{\phi_{Fp}}{kT/q}\right)$$

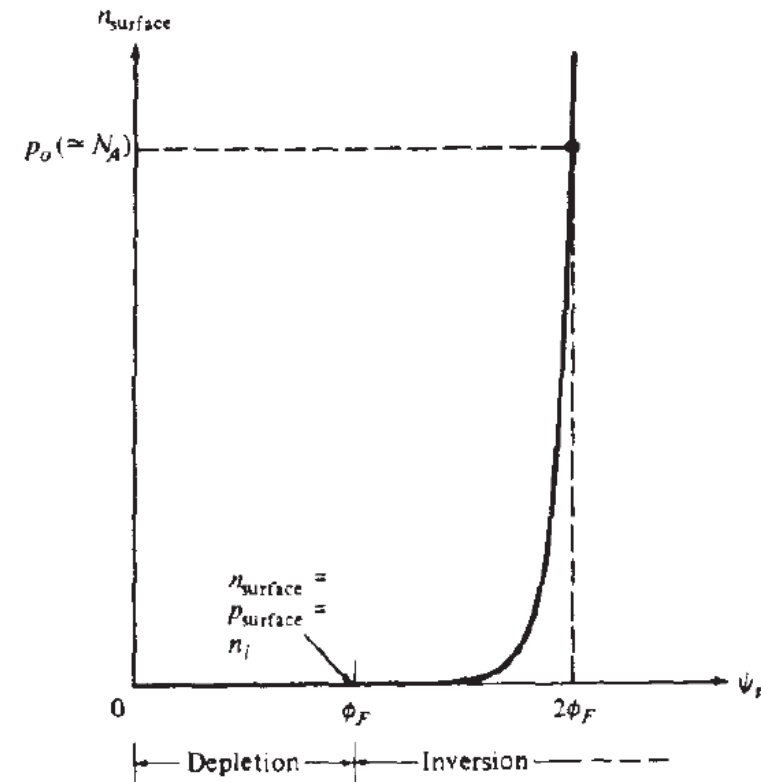
$$2. \quad n_{surface} = n_i \exp\left(-\frac{E_{Fi} - E_F}{kT}\right) = n_i \exp\left(-\frac{\phi_{Fp} - \psi_s}{kT/q}\right)$$

$$\begin{aligned} 3. \quad n_{surface} &= N_A \exp\left(-\frac{\phi_{Fp}}{kT/q}\right) \exp\left(-\frac{\phi_{Fp} - \psi_s}{kT/q}\right) \\ &= N_A \exp\left(\frac{\psi_s - 2\phi_{Fp}}{kT/q}\right) \end{aligned}$$

MOS Capacitor: Surface Potential vs. Electron Concentration at Surface (2)

$$n_{surface} = n_i \exp\left(-\frac{\phi_{Fp} - \psi_s}{kT/q}\right) = N_A \exp\left(\frac{\psi_s - 2\phi_{Fp}}{kT/q}\right)$$

- At $\psi_s = \phi_{Fp}$, $n_{surface} = n_i$ (surface becomes intrinsic)
- This point is the change from depletion to inversion
- When increases more than this point, $n_{surface}$ increases sharply
- At $\psi_s = 2\phi_{Fp}$, $n_{surface} = N_A$ (surface has same carrier concentration as bulk, but with opposite type)
- The surface considered to be strongly inverted



Source: Tsividis, Yannis. *Operation and Modeling of the MOS Transistor*. McGraw-Hill, Inc., 1987.

References

- Tsividis, Yannis. *Operation and Modeling of the MOS Transistor*. McGraw-Hill, Inc., 1987.
- D. A. Neaman, "*Semiconductor Physics and Devices*"